

Calibrated Src biosensor fluorescence trajectories

Scenario and task

Calibrated Src biosensor fluorescence trajectories. Native measurements come from the authoritative source. Complete dose curves; seven development and two hash-selected confirmation doses; leave-one-dose-out validation. No biological replicate identifiers.

Predict all $t > 7.5$ min from source dose and same-curve measurements at 3,4.5,6 min. Calibration fixed at 6 min; no later response access.

Experimental method

8 substantive development attempts tested distinct dynamic, mechanistic or ablation hypotheses. Selection used equally weighted whole-group MAE, with a 1% simpler-model tie rule. Baseline fitting and preparation did not count as attempts. All states and stopping were frozen before a separate confirmation command; confirmation never changed the reference.

First 9 lines and schemas inspected; early values through 6.75 min were visible across doses, outside target window. Later outcomes not printed before freeze.

Reference equation and interpretation

$$\widehat{F}(t) = F_6 + (t - \tau)(av_6 + \tau bc_6), \quad a = 0.882574, \quad b = 0.122161.$$

Let $\tau = 6$ min. Here F is fluorescence in source instrument units, t is minutes, $v_6 = (F_6 - F_{4.5})/(1.5 \text{ min})$ is the calibrated slope, and $c_6 = (F_6 - 2F_{4.5} + F_3)/(2.25 \text{ min}^2)$ is calibrated curvature; a and b are dimensionless. The equation is a predictive tangent correction, not an identified phosphorylation law.

Full-precision coefficients and all comparator states are in `rules.json`.

Findings and performance

The development-selected curvature projection does not confirm a prediction gain. Its held-out MAE is 130.233 source units, versus 126.725 for the unadjusted early tangent, 111.214 for the flexible control and 82.729 for the concentration-based Michaelis-Menten rate control. The lower-error controls were not promoted after confirmation.

Frozen model	Confirmation MAE (source fluorescence units)
reference	130.233
baseline persistence	291.155
baseline linear	126.725
baseline mm	82.729
baseline flexible	111.214

All candidate results and per-group errors remain in `evidence/metrics.csv` and `evidence/by_group.csv`. The reference is the preselected model, not the retrospectively best confirmation model. No error is converted into invented percentage accuracy.

Reserved group	Selected reference MAE
dose-1	95.6238
dose-5	164.843

Value, limits and negative evidence

Nine source dose curves are not nine biological replicates. Source calibration maps intensity to fluorescence units rather than enzyme amount; optical and kinetic effects remain confounded. An Akt 1 CSV response equals its time axis and is excluded; no synthetic identity is scored as discovery. Calibration is permitted for every comparator. No new physical law or clinical impact is established.

Early curvature improved development but not reserved doses. Include a concentration-law control even when it looks weak on development. Optimization boundary contact and independent-dose failures are evidence against mechanistic identification. Never score a copied time axis as biochemical response.

Reproduction and sources

Run `python run.py` inside this final package to verify its manifest and replay every frozen model. `rules.json` defines the exact coefficients, permitted input columns and units. `task_spec.json` and the shared evaluator provide the evaluation contract; valid alternative equations need not resemble these references.

Original study: <https://pubmed.ncbi.nlm.nih.gov/42619562/>. PET-quenching relief under phosphorylation and Michaelis-Menten initial velocities are reported by the authors. Our late-curve calibrated forecast is separate; neither first-order saturation nor power laws are novel families. All packaged outcomes are exposed to future users. Agent review is computational/scientific criticism, not independent experimental replication or proof of novelty.